PCB DESIGN PROJECT REPORT

# LED RUNNING LIGHT CIRCUIT

Sequential LED Chaser using NE555 Timer and CD4017 Counter

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| **Assignment** | Question 12 |
| **Software** | KiCad |
| **Project Type** | Single-sided / two-layer low-voltage PCB concept |

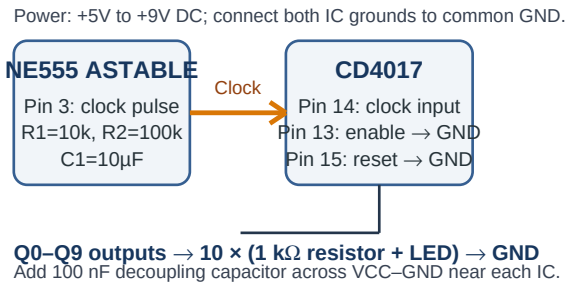
Submitted as a PCB design assignment covering circuit schematic, component footprints, PCB layout, visualisation and bill of materials.

# 1. Project Overview

The LED Running Light Circuit produces a moving light effect by switching ten LEDs on one after another. An NE555 timer is configured in astable mode to generate continuous clock pulses. These pulses drive the clock input of a CD4017 decade counter. The CD4017 activates its outputs sequentially, creating the running-light effect.

# 2. Circuit Schematic

Functional connection diagram (use this as the wiring reference while creating the electrical schematic in KiCad):




## Important pin connections

| IC / Pin | Connection |
|---|---|
| NE555 pin 1 | GND |
| NE555 pin 8 | +5V to +9V DC |
| NE555 pin 4 (RESET) | +VCC |
| NE555 pins 2 and 6 | Join together; timing capacitor C1 to GND |
| NE555 pin 7 | R1 to +VCC; R2 to pins 2/6 |
| NE555 pin 3 | CD4017 pin 14 (CLOCK) |
| CD4017 pin 16 | +VCC |
| CD4017 pin 8 | GND |
| CD4017 pin 13 (CLOCK ENABLE) | GND |
| CD4017 pin 15 (RESET) | GND for continuous 10-step sequence |
| CD4017 pins 3, 2, 4, 7, 10, 1, 5, 6, 9, 11 | Each output to a 1 kΩ resistor, then LED anode; LED cathode to GND |

Note: CD4017 output pin order above follows Q0–Q9: 3, 2, 4, 7, 10, 1, 5, 6, 9, 11. Confirm symbol pin numbering against the selected KiCad library symbol and the exact IC datasheet before fabrication.

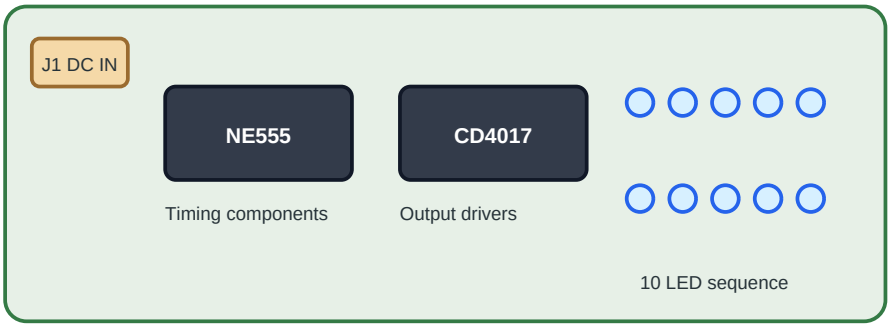
# 3. Component Footprints

Suggested through-hole footprints for a beginner-friendly PCB. Select equivalent footprints available in the installed KiCad libraries and verify pin numbering before assigning them.

| Component | Suggested KiCad footprint |
|---|---|
| NE555 (DIP-8) | Package_DIP:DIP-8_W7.62mm |
| CD4017 (DIP-16) | Package_DIP:DIP-16_W7.62mm |
| LEDs (10) | LED_THT:LED_D3.0mm |
| Resistors (10 + timing) | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal |
| Timing capacitor (10 µF) | Capacitor_THT:CP_Radial_D5.0mm_P2.00mm |
| Decoupling capacitors (100 nF ×2) | Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm |
| DC input connector | Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical |

# 4. PCB Layout — Component Placement and Routing

Place the DC input connector near one board edge. Keep the NE555 timing network close to the NE555 pins 2, 6 and 7. Place the CD4017 near the centre and arrange the ten LEDs in a straight row along the opposite edge. Keep each LED series resistor close to its LED or counter output trace. Use a common ground pour where practical and keep power and ground tracks wide enough for the expected current.




*Conceptual placement only — recreate, route, and verify in KiCad before submission.*

# 5. 2D PCB View

The placement drawing on the previous page is a conceptual guide, not an exported KiCad PCB screenshot. In KiCad PCB Editor, update the PCB from the schematic, arrange footprints, draw the board outline on Edge.Cuts, route all nets, run Design Rules Checker (DRC), and capture the final 2D view.

# 6. 3D PCB View

Open View → 3D Viewer in KiCad after assigning footprints and completing the board. Capture a screenshot showing the board, connectors, both IC packages and all ten LEDs. The 3D view must come from your actual KiCad project; a conceptual illustration is not a substitute.

# 7. Bill of Materials (BOM)

| Ref. | Part / value | Qty | Notes |
|---|---|---|---|
| U1 | NE555 timer, DIP-8 | 1 | Astable clock generator |
| U2 | CD4017 decade counter, DIP-16 | 1 | 10-step sequence |
| D1–D10 | LED, 3 mm or 5 mm | 10 | One LED per counter output |
| R3–R12 | 1 kΩ resistor | 10 | LED current limiting |
| R1 | 10 kΩ resistor | 1 | 555 timing resistor |
| R2 | 100 kΩ resistor | 1 | 555 timing resistor |
| C1 | 10 µF electrolytic | 1 | 555 timing capacitor |
| C2–C3 | 100 nF ceramic | 2 | IC supply decoupling |
| J1 | 2-pin DC input connector | 1 | 5–9 V DC input |

# 8. Working Principle and Design Notes

The NE555 charges and discharges C1 through R1 and R2, producing a periodic square-wave signal at pin 3. Each rising clock edge advances the CD4017 by one output. The active output lights its associated LED through a current-limiting resistor. After Q9, the counter returns to Q0 and repeats.

Approximate 555 astable frequency: $f \approx 1.44 / ((R1 + 2R2) \times C1)$. With R1 = 10 kΩ, R2 = 100 kΩ and C1 = 10 µF, the nominal frequency is about 0.69 Hz. Actual speed varies with component tolerances. Use a regulated 5 V supply if using a 5 V CD4017 variant; verify the exact IC part ratings and LED current before powering the board.

# 9. KiCad Completion and Export Checklist

- Create the project and draw the schematic using NE555, CD4017, LED, resistor, capacitor and connector symbols.
- Annotate, run Electrical Rules Checker (ERC), and assign the suggested footprints.
- Open PCB Editor, update PCB from schematic, draw Edge.Cuts outline, place components and route all nets.
- Add ground zone if appropriate, refill zones, run DRC and resolve all errors.
- Capture actual screenshots: schematic, footprint/PCB after routing, 2D PCB view and 3D Viewer.
- Plot Gerber files (copper, mask, silkscreen and Edge.Cuts) and generate drill files from the completed PCB.

### Important submission note

This report provides the circuit design reference, suggested footprints, layout guidance and BOM. The actual KiCad schematic/PCB screenshots and Gerber manufacturing files must be generated from a completed, electrically checked KiCad project. Do not present the conceptual placement illustration as a routed PCB or 3D Viewer screenshot.